

Vikas Kumar

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Vikas-eng-aiml

LinkedIn

PROFESSIONAL SUMMARY

B.Tech Computer Science and Engineering graduate specializing in Artificial Intelligence and Machine Learning, with strong foundations in Full Stack Development, AI/ML, and Generative AI. Hands-on experience with React.js, Node.js, Express.js, MongoDB, SQL, Python, and JavaScript through academic and personal projects. Passionate about building scalable web applications and intelligent solutions using modern software development, machine learning, and AI technologies.

EDUCATION

Vellore Institute of Technology | B.Tech

B.Tech in Computer Science and Technology

CGPA: 8.58/10

2023–2027

PR + 2 High School Korahia Jaynagar | 12th Standard

Percentage: 85.8%

2021–2022

PR High School Korahia Jaynagar | 10th Standard

Percentage: 81.2%

2019–2020

TECHNICAL SKILLS

- **Languages:** C++ (Proficient), Python, JavaScript, SQL
- **Frontend:** HTML5, CSS3, JSX, Flexbox, Grid, Responsive Design, **React.js**, Tailwind CSS, Bootstrap, Material UI
- **Backend & REST APIs:** Node.js, Express.js, RESTful API Design & Development, MVC Architecture, Middleware, Authentication & Authorization
- **Databases:** MongoDB (CRUD, Aggregation, Indexing), SQL Databases
- **AI / GenAI:** LLMs, Prompt Engineering, OpenAI & Gemini APIs, **RAG Pipelines**, LangChain, LangGraph

PROJECTS

• Invisible Watermarking Tool — Image Security App



React.js, Flask, Python, OpenCV, DCT

- Built a web-based application for embedding and extracting invisible image watermarks using DCT-based frequency-domain techniques.
- Implemented configurable watermark strength and robust extraction mechanisms.
- Established watermark extraction methods that recovered embedded data while preserving high visual quality, achieving **99.5% accuracy** across all test images.

• RF-Based Non-Invasive Glucose Monitoring with Machine Learning Models



Python, Machine Learning, Scikit-learn, XGBoost, Microstrip Antenna, RF Simulation

- Engineered a state-of-the-art glucose monitoring solution leveraging machine learning algorithms; reduced prediction error margin by 20% through rigorous testing and iterative adjustments to improve user experience in clinical settings.
- Key RF features including resonant frequency shift, S11, impedance, dielectric constant, and loss tangent were extracted for model training.
- Designed and executed a hybrid predictive analytics framework leveraging Random Forest and XGBoost; results demonstrated improved regression outcomes, achieving a **20% reduction in prediction error** on key performance metrics.

CERTIFICATIONS & ACHIEVEMENTS

- Generative AI Professional Certificate — Oracle
- Agentic AI with LangChain and LangGraph — IBM
- AWS Technical Essentials Certificate — AWS
- Advanced RAG with Vector Databases and Retrievers — IBM
- Machine Learning with Python Certificate — Udemy

EXTRA-CURRICULAR ACTIVITIES

- **National Service Scheme (NSS) Volunteer:** Executed a tree plantation drive resulting in the successful planting of **1,200 trees** in partnership with local schools; promoted environmental stewardship among students, leading to lasting community involvement in sustainability efforts.
- **Event Coordination — University Convocation:** Member of the organizing team for a convocation ceremony attended by the Chief Minister of Madhya Pradesh.
- **Technical Team Member — Notion Community Club:** Contributed to building and managing Notion workspaces and technical documentation.

ADDITIONAL INFORMATION

- **Languages:** Hindi (Native), Maithili (Native), English (Fluent)